



The neurobiology of depression, ketamine and rapid-acting antidepressants: Is it glutamate inhibition or activation?

Chadi G. Abdallah^{a,b,*}, Gerard Sanacora^{a,c}, Ronald S. Duman^{a,b,c}, John H. Krystal^{a,b,c}

^a Department of Psychiatry, Yale University School of Medicine, New Haven, USA

^b Clinical Neuroscience Division, Department of Veterans Affairs National Center for Posttraumatic Stress Disorder, Veterans Affairs Connecticut Healthcare System, West Haven, USA

^c Abraham Ribicoff Research Facilities, Connecticut Mental Health Center, New Haven, USA

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ABSTRACT

The discovery of the antidepressant effects of ketamine has opened a breakthrough opportunity to develop a truly novel class of safe, effective, and rapid-acting antidepressants (RAADs). In addition, the rapid and robust biological and behavioral effects of ketamine offered a unique opportunity to utilize the drug as a tool to thoroughly investigate the neurobiology of stress and depression in animals, and to develop sensitive and reproducible biomarkers in humans. The ketamine literature over the past two decades has considerably enriched our understanding of the mechanisms underlying chronic stress, depression, and RAADs. However, considering the complexity of the pharmacokinetics and in vivo pharmacodynamics of ketamine, several questions remain unanswered and, at times, even answered questions continue to be considered controversial or at least not fully understood. The current perspective paper summarizes our understanding of the neurobiology of depression, and the mechanisms of action of ketamine and other RAADs. The review focuses on the role of glutamate neurotransmission – reviewing the history of the “glutamate inhibition” and “glutamate activation” hypotheses, proposing a synaptic connectivity model of chronic stress pathology, and describing the mechanism of action of ketamine. It will also summarize the clinical efficacy findings of putative RAADs, present relevant human biomarker findings, and discuss current challenges and future directions.

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Abbreviations: ¹³C MRS, carbon-13 magnetic resonance spectroscopy; ABP, aminoacid-based pathology; AMPAR, α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor; BDNF, brain derived neurotrophic factor; BOLD, blood oxygen level dependent; CSP, chronic stress pathology; GAD, generalized anxiety disorder; GBC, global brain connectivity; GluR2B, N-methyl-D-aspartate receptor subtype 2B; HNK, hydroxynorketamine; i.n., intranasal; i.v., intravenous; M-AChR, muscarinic cholinergic receptor; MBP, monoamine-based pathology; mGluR5, metabotropic glutamate receptor subtype 5; MRI, magnetic resonance imaging; mTROC1, mechanistic target of rapamycin complex 1; NAc, nucleus accumbens; NMDAR, N-methyl-D-aspartate receptor; OCD, obsessive compulsive disorder; PFC, prefrontal cortex; PTSD, posttraumatic stress disorder; RAAD, rapid-acting antidepressant; SAAD, slow-acting antidepressant; SAD, social anxiety disorder; SRI, serotonin re-uptake inhibition; VDCC, L-type voltage-dependent calcium channels.

* Corresponding author at: Clinical Neuroscience Division, National Center for PTSD, 950 Campbell Ave 151E, West Haven, CT 06516, USA.

E-mail address: chadi.abdallah@yale.edu (C.G. Abdallah).

1. Introduction

Serendipity, combined with astute clinical observations, has dominated the path to drug discovery in psychiatry (Klein, 2008). In 1951, the first antipsychotic drug was discovered unexpectedly, as chlorpromazine was being developed for potentiating anesthesia. The first tricyclic antidepressant, imipramine, was synthesized in 1899 and decades later failed as antipsychotic compound (Ban, 2006). Yet, one case report in mid-1950s showing imipramine's antidepressant effect in a female with severe depression has led to further investigation and eventual discovery of the monoaminergic class of antidepressants. Similarly, the first benzodiazepine was lingering on a laboratory shelf for years until it was accidentally discovered during a "spring-cleaning" in 1957 and subsequently demonstrated strong anxiolytic effects (Ban, 2006).

Another unanticipated observation in the 1950s was the report that the anti-tuberculosis D-cycloserine, an *N*-methyl-D-aspartate receptor (NMDAR) modulator, may possess antidepressant properties (Crane, 1959). Yet, this fortuitous observation has gained little to no attention for more than four decades, until it was discovered in the late 1990s that a single subanesthetic dose of the NMDAR antagonist ketamine induces rapid and sustained antidepressant effects in severely depressed patients (Berman et al., 2000). At the time, in the context of accumulating evidence proposing NMDAR modulation as a target for antidepressants, and relating depression to excess glutamate neurotransmission and excitotoxicity, the ketamine findings have generated considerable interest in the field to target glutamate neurotransmission for the development of novel rapid-acting antidepressants (RAADs) (Berman et al., 2000; McEwen, 1999; Skolnick et al., 1996; Zarate et al., 2006). Early attempts have primarily focused on investigating glutamate release inhibitors and NMDAR antagonists, both of which were thought to inhibit glutamate transmission and offset the depression-related excitotoxicity. Unfortunately, the glutamate release inhibition approach has had limited success in human studies over the past 2 decades, with pilot or inconsistent findings of antidepressant properties following sustained treatment and no evidence of RAAD effects (Mathew, Gueorguieva, Brandt, Fava, & Sanacora, 2017; Solmi et al., 2016). Conversely, the NMDAR antagonism approach has shown promise (Abdallah, Averill, & Krystal, 2015; Bobo et al., 2016). Yet, it is becoming increasingly apparent that the NMDAR agents with RAAD properties are putatively exerting their effects through glutamate neurotransmission activation, rather than inhibition (Aleksandrova, Wang, & Phillips, 2017; Murrough, Abdallah, & Mathew, 2017).

In this perspective paper, we will (1) review the history of the "glutamate inhibition" and "glutamate activation" hypotheses, (2) propose a synaptic connectivity model of chronic stress pathology, (3) describe the mechanism of action of ketamine, (4) summarize the clinical efficacy findings of putative RAADs, (5) present relevant human biomarker findings, and (6) discuss current challenges and future directions.

2. Glutamate inhibition or activation? A historical perspective

Early in the 1990s, a number of NMDAR antagonists have demonstrated antidepressant-like effects in rodents (Trullas & Skolnick, 1990). Follow-up studies have later shown that chronic, but not acute, administration of several traditional antidepressants (i.e., slow-acting antidepressants; SAADs) alter NMDAR binding, leading to the hypothesis that downregulation of NMDAR function may be a common pathway across antidepressants (Skolnick et al., 1996). During the same period, gray matter structural deficits were demonstrated in stress-related disorders in humans (Bremner et al., 1995; Sheline, Wang, Gado, Csernansky, & Vannier, 1996), and were thought to parallel the dendritic atrophy observed following chronic stress in rodents (McEwen, 1999). Interestingly, inhibiting NMDARs or glutamate release blocked the effects of chronic stress on dendritic atrophy

(McEwen, 1999). Together, these early data supported a model in which downregulation of excess glutamate may exert antidepressant effects, and raised the question whether the RAAD effects of ketamine are due to glutamate neurotransmission inhibition by blocking NMDARs.

In contrast to the glutamate inhibition model, it has been previously shown that subanesthetic doses of ketamine transiently activate rather than inhibit glutamate neurotransmission (Moghaddam, Adams, Verma, & Daly, 1997). Moreover, the 1990s also witnessed the rise of the neurotrophic hypothesis of depression (Duman, Heninger, & Nestler, 1997), which associated chronic stress and depression with a deficit in brain derived neurotrophic factor (BDNF) and demonstrated that traditional antidepressants increase brain BDNF expression (Nibuya, Morinobu, & Duman, 1995). Interestingly, acute glutamate neurotransmission activation – rather than inhibition – was initially associated with upregulation of BDNF and other neurotrophics (Gall & Isackson, 1989; Patterson, Grover, Schwartzkroin, & Bothwell, 1992; Zafra, Castren, Thoenen, & Lindholm, 1991). In addition, it was found that glutamate transmission activation, using α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA) potentiators, increases brain BDNF and exhibits RAAD properties in rodent models (Lauterborn, Lynch, Vanderklisch, Arai, & Gall, 2000; Li et al., 2001). Hence, by early 2000s, convergent evidence strongly supported the role of neuronal plasticity in the pathophysiology of depression and in the mechanisms of antidepressant action (D'Sa & Duman, 2002; Manji, Drevets, & Charney, 2001; McEwen, 2004). In addition, it became evident that targeting glutamate neurotransmission offers a novel approach for discovery of new antidepressants (Javitt, 2004; Krystal et al., 2002). However, it was not fully clear whether these novel antidepressants should activate and/or inhibit glutamate neurotransmission, with the latter possibility gaining the most attention considering the evidence available at the time.

3. Synaptic model of chronic stress pathology (CSP)

The synaptic CSP model proposes that trauma and repeated stressors lead to wide spread neuronal remodeling consistent with both reduced and increased synaptic connectivity, depending on the brain region. The chronic stress induced reduction in synaptic connectivity has been mostly studied in the prefrontal cortex (PFC) and the hippocampus. Conversely, the CSP-related increases in synaptic connectivity were most commonly shown in the nucleus accumbens (NAc) and certain nuclei within the amygdala.

In the PFC (Fig. 1), it was shown that prolonged stress precipitates neuronal synaptic hypoconnectivity, as evident by reduced dendritic length and arborization, and by reduction in synaptic density and strengths (Duman & Aghajanian, 2012). Glial cells, which play a critical role in regulating glutamate neurotransmission and preventing excitotoxicity, were also found to be deficient following chronic stress (Sanacora & Banasr, 2013). While the mechanisms underlying the CSP-related hypoconnectivity are not fully known, accumulating evidence implicates glucocorticoid signaling and dysregulation in glutamate neurotransmission (Popoli, Yan, McEwen, & Sanacora, 2012; Sanacora, Treccani, & Popoli, 2012). In particular, trauma- and stress-related dysregulation of glucocorticoid signaling and glutamate release, combined with glial deficit and reduced glutamate uptake, are believed to paradoxically maintain high levels of extracellular glutamate despite the reduction of resting prefrontal synaptic glutamate neurotransmission following chronic stress.

In this model, acute stress precipitates a prefrontal glutamate surge associated with transient (minutes-to-hours) increase in extracellular glutamate (Moghaddam, 1993), but sustained (days-to-weeks) increase in NMDARs, AMPARs, and synaptic strength (Yuen et al., 2009; Yuen et al., 2011). In contrast, chronic stress leads to a sustained increase in extracellular glutamate (Li et al., 2018), combined with reduced resting prefrontal glutamate transmission (Banasr et al., 2010),

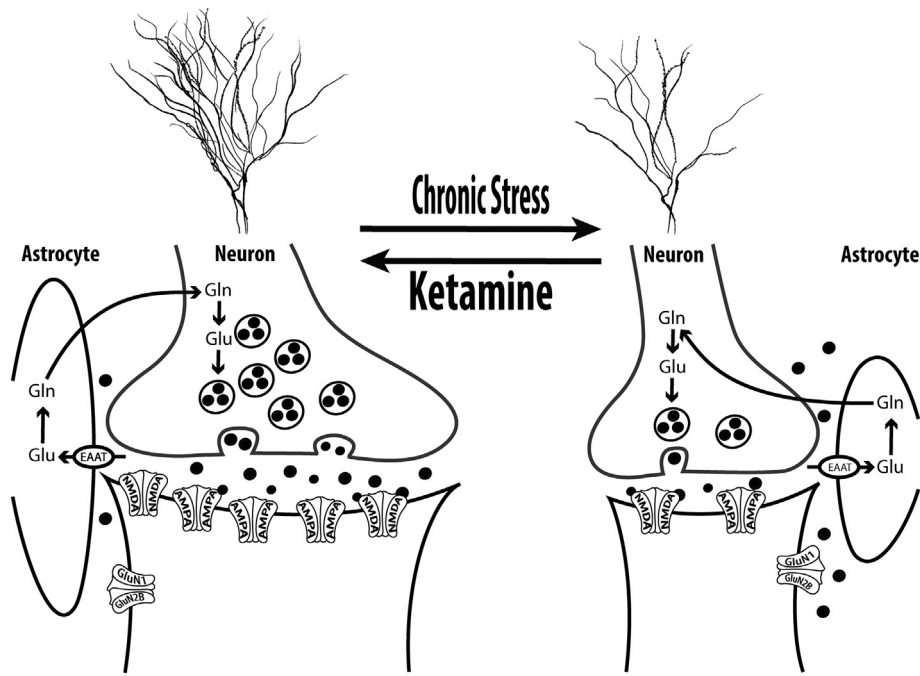


Fig. 1. Chronic Stress Pathology (CSP) in the Prefrontal Cortex (PFC). The synaptic CSP model proposes that synaptic dysconnectivity may be a common pathological pathway across psychiatric disorders with chronic stress component – as a predisposition, a trigger, or an outcome. In the PFC, chronic stress is believed to induce glial deficit, leading to reduced glutamate reuptake capacity and increased extrasynaptic glutamate levels and excitotoxicity. Subsequently, neuronal atrophy develops, resulting in overall reduction in glutamate neurotransmission, which reflects reduced dendritic length and branching, and reduction of spines and synapses density. In the remaining PFC synapses, the neurotransmission strength is also affected by reduced postsynaptic glutamate N-methyl-D-aspartate (NMDA) and α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA) receptors. Ketamine reverses this PFC CSP within 24 h of injection. It is thought that ketamine induces a transient (minutes-to-hours) postsynaptic glutamate activation, which leads to upregulation of neurotrophic signaling, increased protein synthesis, and sustained (days-to-weeks) restoration of synaptic connectivity. Abbreviations: EAAT = excitatory aminoacid transporter; Gln = glutamine; GluN1 = NMDA subtype 1; GluN2B = NMDA subtype 2B; Glu = glutamate. The figure was adapted with permission from the Emerge Research Program (emerge.care).

and reduction in NMDARs, AMPARs, and synaptic strength (Yuen et al., 2012). Here, the distinction between “acute” and “chronic” stress is critical, with the timing (i.e., acute vs. chronic) pertains mostly to the length of the stress response – rather than the duration of the stressor. For example, a single severe traumatic event may induce a chronic sustained threat response. Conversely, repeated escapable and/or predictable mild stressors will result in appropriate adaptation with only acute transient stress responses.

In the NAc, a number of chronic stress paradigms were found to increase synaptic connectivity, as evident by increased dendritic length and arborization, as well as increased synaptic density and strength (Campioni, Xu, & McGehee, 2009; Christoffel et al., 2011; Christoffel et al., 2012; Coplan et al., 2018; Muhammad, Carroll, & Kolb, 2012; Warren et al., 2014). While the prefrontal hypoconnectivity was associated with glutamate dysregulation and excitotoxicity, the stress-induced NAc synaptic hyperconnectivity is related to monoamine dysregulation. In particular, chronic stress leads to phasic activation of the dopaminergic neurons from the ventral tegmental area (Chaudhury et al., 2013), which precipitates the co-release of dopamine and BDNF in the NAc (Walsh et al., 2014). Subsequently, the BDNF upregulation and the induction of its high affinity receptor TrkB lead to the CSP-related NAc neuronal hypertrophy (Wook Koo et al., 2016).

In preclinical studies, depressive-like behaviors were directly associated with these synaptic alterations in the PFC and NAc (Duman, Aghajanian, Sanacora, & Krystal, 2016; Krishnan & Nestler, 2008; Russo & Nestler, 2013). Reversal of the synaptic impairment induces antidepressant effects. Moreover, both SAADs and RAADs are known to increase PFC, but reduce NAc, synaptic connectivity (Hare, Ghosal, & Duman, 2017; Melo et al., 2015; Yao, Skiteva, Zhang, Svenningsson, & Chergui, 2017). Notably, the CSP-related microstructural synaptic alterations are evident at the macrostructural level as assessed by magnetic resonance imaging (MRI) (Kassem et al., 2013). Thus, providing

support for the synaptic CSP model, human MRI studies have shown increased NAc, but reduced hippocampal and PFC volumes in major depression (Abdallah et al., 2017; Kempton et al., 2011). Here, it is important to highlight that the PFC and hippocampal gray matter deficits were absent in several human depression studies. These gray matter deficits are most evident in patients with a reduction of amino acid neurotransmitters (i.e., glutamate & GABA) and in individuals who are treatment resistant to SAADs, which are primarily monoaminergic drugs [reviewed in (Abdallah et al., 2015; Abdallah, Jackowski, Salas, et al., 2017)]. Therefore, it was proposed that the synaptic PFC/hippocampus hypoconnectivity and NAc hyperconnectivity reflect two pathways that may independently precipitate clinical depression (Abdallah, Jackowski, Salas, et al., 2017). In this Dual Pathology model, patients with underlying amino acid-based pathology (ABP), leading to excitotoxicity and synaptic loss, would show PFC/hippocampus gray matter deficit, present with amino acid impairment, and be treatment resistant to monoaminergic antidepressants. Conversely, patients with monoamine-based pathology (MBP), leading to localized increase in BDNF and synaptic gain, would show NAc gray matter hypertrophy, lack of amino acid impairment, and effectively respond to monoaminergic antidepressants (Abdallah, Jackowski, Salas, et al., 2017).

Finally, although the synaptic CSP model has been typically studied and interpreted within the context of major depression, CSP appears to be a common pathway across numerous psychiatric disorders (Adams et al., 2018; Daskalakis & Binder, 2015; Goddard, 2017; Kwako & Koob, 2017; Maeng & Milad, 2017; Patriquin & Mathew, 2017; Prescott et al., 2018). Hence, the evidence of synaptic loss and dysconnectivity is not limited to major depression, but rather common to several stress-related disorders – e.g., posttraumatic stress disorder (PTSD), generalized anxiety disorder (GAD), obsessive compulsive disorder (OCD), and bipolar depression (Abdallah et al., 2013, 2017; Akiki et al., 2017; Anticevic et al., 2013; Anticevic et al., 2014;

Averill et al., 2017; Haukvik et al., 2015; Kwon et al., 2003; Pietrzak et al., 2015; Wrocklage et al., 2017). In addition, antidepressants – known to reverse the CSP in animals – have shown efficacy in alleviating symptoms of PTSD, GAD, OCD, bipolar depression, and other disorders with a considerable chronic stress component. Here, we note that synaptic dysconnectivity could be a predisposing factor, an outcome, or a perpetrator of the psychopathology (Averill et al., 2017; Matosin, Cruceanu, & Binder, 2017; Sheth, McGlade, & Yurgelun-Todd, 2017; Syed & Nemeroff, 2017). Furthermore, while synaptic loss appears to be common across stress-related disorders, the location and pattern of the synaptic dysconnectivity, combined with individual characteristics (e.g., genes & environment), may be the mechanism through which CSP is associated with distinct clinical presentations and psychopathologies (Abdallah, Southwick, & Krystal, 2017; Averill, Purohit, et al., 2017; Krystal et al., 2017). Together, the presented synaptic model proposes that CSP is common across many psychiatric disorders and that targeting synaptic connectivity may be a convergent pathway across antidepressants.

4. Mechanism of action of ketamine and RAADs

The discovery of the robust RAAD effects of ketamine offered a unique opportunity to better understand the neurobiology of depression and to unravel the processes involved in reversing CSP. To date, two ketamine-induced glutamate neurotransmission changes appear to be critical to its RAAD effects: (1) a transient activation of glutamate neurotransmission in the PFC (often referred to as glutamate “surge” or “burst”) and (2) a sustained increase in PFC synaptic connectivity (Fig. 2). It is believed that acute administration of subanesthetic doses of ketamine induces a transient surge in prefrontal glutamate neurotransmission, which in turn induces a number of intracellular

processes ultimately leading to sustained increase in prefrontal synaptic connectivity within 24 h of treatment (Abdallah et al., 2016; Duman et al., 2016).

Transient PFC glutamate neurotransmission activation leads to activity-dependent release of BDNF, activates the mechanistic target of rapamycin complex 1 (mTORC1) signaling, and increases protein synthesis and synaptic strength (Lepack, Bang, Lee, Dwyer, & Duman, 2016; Lepack, Fuchikami, Dwyer, Banasr, & Duman, 2015; Liu et al., 2012). The preclinical evidence of ketamine-induced glutamate release has long been demonstrated using microdialysis (Moghaddam et al., 1997) and, more recently, evidence of transient glutamate transmission activation was demonstrated using ex vivo carbon-13 magnetic resonance spectroscopy (^{13}C MRS) (Chowdhury et al., 2012; Chowdhury et al., 2017; Rothman, De Feyter, de Graaf, Mason, & Behar, 2011). Here, it is important to distinguish between presynaptic glutamate release and postsynaptic activation. The latter – i.e., postsynaptic glutamate activation – is required for the induction of BDNF and its high affinity receptor TrkB, the increase in synaptic strength, and the RAAD effects. Hence, mechanisms independent of evoked presynaptic release of glutamate have been proposed for ketamine (Autry et al., 2011; Miller, Moran, & Hall, 2016) and agents directly targeting postsynaptic glutamate activation may possess RAAD properties (e.g., AMPAR potentiators or NMDAR partial agonists; Fig. 2).

The role of the transient postsynaptic glutamate activation in the RAAD effects of ketamine has been abundantly shown in preclinical studies. First line of evidence stems from a consistent observation that the inhibition of postsynaptic glutamate activation – using AMPAR antagonists – blocks the synaptic remodeling and the RAAD effects of ketamine (Li et al., 2010; Maeng et al., 2008). Here, we note that even selective AMPAR antagonists will presumably block overall postsynaptic glutamate activation through inhibition of both AMPAR and

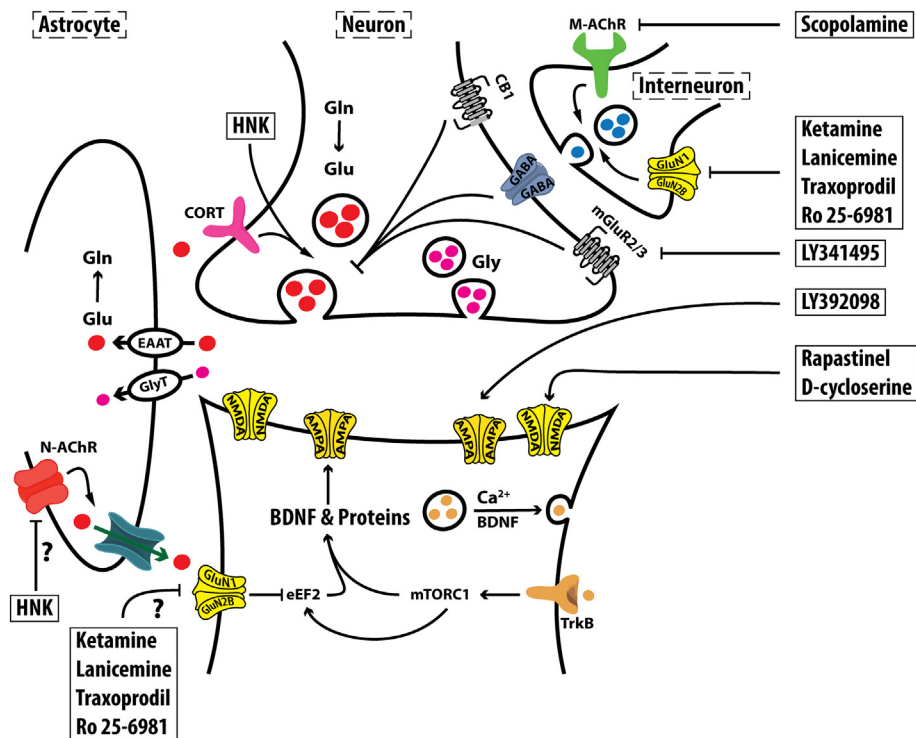


Fig. 2. Molecular Targets of Rapid-acting Antidepressants (RAADs). It is believed that RAADs exert their effects by inducing a transient (minutes-to-hours) postsynaptic glutamate activation, which ultimately leads to sustained (days-to-weeks) increase in synaptic formation and strength in the prefrontal cortex. It remains to be determined in future studies whether inhibition of extrasynaptic N-methyl-D-aspartate (NMDA) receptors would be sufficient to exert RAAD effects. The figure depicts the potential targets of agents suspected to have RAAD properties. Abbreviations: AMPA = α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor; BDNF = brain derived neurotrophic factor; CB1 = cannabinoid receptor; EAAT = excitatory amino acid transporter; eEF2 = eukaryotic elongation factor 2; Gln = glutamine; GluN1 = NMDA subtype 1; GluN2B = NMDA subtype 2B; Glu = glutamate; Gly = Glycine; GlyT = Glycine transporter; HNK = hydroxynorketamine; M-AChR = muscarinic acetylcholine receptor; mGluR2/3 = metabotropic glutamate receptor subtype 2 and 3; mTORC1 = mechanistic target of rapamycin complex 1; N-AChR = nicotinic AChR; TrkB = tyrosine kinase B receptor. The figure was adapted with permission from the Emerge Research Program (emerge.care).

NMDAR, considering that AMPAR activation leading to membrane depolarization is required for NMDAR signaling. In this context, most of the evidence relating AMPAR blockade to the inhibition of RAADs would be considered an indication of the need for postsynaptic glutamate activation rather than specific AMPAR activation. For example, NMDAR potentiators – without AMPAR modulating properties – may possess RAAD effects by inducing transient postsynaptic glutamate activity (e.g., NMDAR partial agonists). Furthermore, the RAAD effects of these NMDAR potentiators will still be blocked by pretreatment with AMPAR antagonists, because the latter also inhibit NMDAR signaling. Another line of evidence underscoring the role of transient glutamate activation is that postsynaptic depolarization and activation of L-type voltage-dependent calcium channels (VDCC) are necessary for the synaptic changes and the RAAD effects of ketamine (Jourdi et al., 2009; Lepack et al., 2015) and other RAADs (Ghosal et al., 2018).

While the mechanisms through which ketamine induces a transient postsynaptic glutamate activation are not fully known, a leading hypothesis is that subanesthetic doses of ketamine preferentially inhibits NMDARs on a subpopulation of interneurons precipitating pyramidal neuron disinhibition and a paradoxical surge in glutamate release (Homayoun & Moghaddam, 2007). However, alternative hypotheses have also been proposed. One study hypothesized that blockade of at rest NMDAR signaling – i.e., without evoked glutamate release – would be sufficient to increase eukaryotic elongation factor 2 (eEF2) signaling and BDNF translation, leading to increased protein synthesis and synaptic connectivity (Autry et al., 2011). Yet, using AMPAR modulation, the same study has also demonstrated that postsynaptic glutamate activation is necessary for the RAAD effects of ketamine (Autry et al., 2011). A main metabolite of ketamine is (2S,6S;2R,6R)-hydroxynorketamine (HNK) (Abdallah, 2017). Following injection, ketamine rapidly reaches the brain within 1 min and maintains a brain/plasma concentration ratio equal to 6.5 for 10 min (Cohen, Chan, Way, & Trevor, 1973). In rodents, the plasma concentrations of ketamine and norketamine, both potent NMDAR antagonists, peak at 10 min post-injection and decrease exponentially thereafter, while the HNK concentration peaks at 30 min (Can et al., 2016; Moaddel et al., 2015; Paul et al., 2014). Recently, it was found that HNK possesses RAAD properties without blockade of NMDARs (Zanos et al., 2016). However, the same study has also demonstrated that postsynaptic glutamate activation is necessary for the RAAD effects of ketamine (Zanos et al., 2016), and preliminary evidence has since shown that (2R,6R)-HNK also induces a glutamate release surge (Pham et al., 2017; Schwarcz, Wu, Zanos, & Gould, 2017). Together, the data highlights the ability of ketamine to induce a surge in glutamate transmission and that the transient postsynaptic activation is responsible for its RAAD effects (Fig. 2).

The role of transient postsynaptic glutamate activation is not limited to the neurobiology of ketamine. In fact, the mechanisms of several other RAADs have been related to transient glutamate activation. Scopolamine, a muscarinic cholinergic receptor (M-AChR) antagonist, was shown to increase glutamate release and postsynaptic activation, leading to increased PFC BDNF and synaptic connectivity, and to RAAD effects. The molecular and behavioral effects of scopolamine are blocked by inhibiting postsynaptic glutamate activation (Chowdhury et al., 2017; Ghosal et al., 2018; Voleti et al., 2013; Wohleb et al., 2016). Similarly, rapastinel (also known as GLYX-13) – a drug with presumable NMDAR partial agonist properties – was shown to increase PFC BDNF and synaptic connectivity, and to exert RAAD effects, all of which were dependent on postsynaptic glutamate activation as evident by manipulation of AMPAR and VDCC (Lepack et al., 2016; Liu et al., 2017). The transient glutamate activation was also related to many other putative RAADs, including the selective NMDAR subtype 2B (GluN2B) antagonists Ro 25-6981 and traxoprodil (also known as CP-101,606), the mGluR2/3 antagonist LY341495, and the AMPAR potentiator LY392098 [*positive allosteric modulators (PAM) or ampakines*] (Chowdhury et al., 2017; Karasawa, Shimazaki, Kawashima, & Chaki, 2005; Li et al., 2001; Tang et al., 2018). Here, it is

important to note that the reviewed literature is specific to transient glutamate effects, and may not necessarily translate to the effects of sustained increase in glutamate activation. Indeed, comparable to CSP, it is a concern that frequent daily administration of ketamine or chronic activation of glutamate may lead to excitotoxicity and synaptic dysconnectivity. Consistent with this concern, extensive preclinical literature relates repeated ketamine administration to neurotoxicity and behavioral abnormalities [e.g., (Schobel et al., 2013)]. Similarly, the substance abuse literature of daily use of ketamine underscores its detrimental effects on cognition and mood (Morgan, Curran, & Independent Scientific Committee on Drugs, 2012). Future studies should investigate whether infrequent glutamate activation, such as twice per week administration of ketamine, would provide optimal balance for maintaining the beneficial synaptic connectivity changes.

While there is evidence to support the SAAD properties of glutamate release inhibitors [e.g., lamotrigine (Solmi et al., 2016)], these medications do not typically induce RAAD effects. Moreover, the ultimate effects of the chronic administration of these glutamate modulators may still be increasing glutamate neurotransmission and BDNF, and subsequent normalization of synaptic connectivity. For example, chronic treatment with the glutamate release inhibitor lamotrigine was shown to reverse the pathology of chronic stress and to increase prefrontal and hippocampal BDNF (Li et al., 2011). In addition, chronic riluzole treatment – an agent believed to inhibit the calcium-dependent glutamate release and increase astrocytic glutamate re-uptake – was shown to reverse CSP and increase overall PFC glutamate neurotransmission activation, rather than decreasing it (Banasr et al., 2010; Chowdhury et al., 2008). Finally, in contrast to synaptic NMDARs where activation would lead to increased synaptic formation and strength (i.e., synaptogenesis), the activation of extrasynaptic NMDARs is thought to promote synaptic death (Hardingham & Bading, 2010). Extrasynaptic NMDARs are activated by excessive extracellular levels of glutamate, which causes overstimulation of NMDARs. This leads to an increased calcium influx, activates toxic metabolic processes and triggers cell death (Deutschenbaur et al., 2016; Paoletti, Bellone, & Zhou, 2013). Therefore, selective blockade of extrasynaptic NMDARs may induce synaptogenesis and exert antidepressant effects. A recent study has shown that targeting extrasynaptic NMDARs would exert RAAD effects in rodents (Li et al., 2018). Another study has shown that ketamine blockade of bursting activities in the lateral habenula precipitates RAAD effects (Yang et al., 2018). However, a major limitation of these studies is that the behavioral effects were tested immediately after ketamine administration, rather than 24 h later to confirm the presence of RAAD effects in the absence of ketamine intoxication (Li et al., 2018; Yang et al., 2018). Additionally, the selective NMDAR modulation approaches used may have inadvertently induced a paradoxical glutamate surge in the PFC, similar to the *in vivo* effects of ketamine and many other NMDAR modulators. Future studies would be necessary to demonstrate the RAAD effects at 24 h post administration, and to determine whether selective blockade of extrasynaptic NMDAR signaling is sufficient to exert RAAD effects without the need for postsynaptic glutamate neurotransmission activation.

5. Clinical efficacy of RAADs

Following the regimen used in the first study (Berman et al., 2000), clinical trials have mostly administered 0.5 mg/kg intravenous (i.v.) ketamine infused over 40 min [reviewed in (Abdallah, Averill, & Krystal, 2015)]. To date, there is well replicated evidence showing the RAAD effects of a single ketamine infusion in MDD (McGirr et al., 2015). Concerns regarding the efficacy of the treatment blinding were partially addressed using active placebo (Murrough et al., 2013). A major limitation of the single infusion treatment is that patients often relapse within 1–2 weeks. However, repeated administration of ketamine (e.g., twice per week) appears to maintain the RAAD effects (Singh et al., 2016). While the need for intravenous administration

could be a limiting factor, pilot evidence suggests that intranasal (i.n.) administration of ketamine may exert RAAD effects (Canuso et al., 2018; Daly et al., 2018; Lapidus et al., 2014). The psychotomimetic effects of ketamine could be considered a limitation, although these adverse events are transient (1–2h) and typically well tolerated. The main remaining limitations of ketamine treatment are its addiction liability and the scarcity of data regarding the safety of chronic treatment (Kokkinou, Ashok, & Howes, 2018; Sanacora, Frye, et al., 2017). The latter is particularly important considering the association of heavy daily use of ketamine with ulcerative cystitis, hepatotoxicity, and neurotoxicity (Cottrell et al., 2008; Morgan, et al., 2012; Noppers et al., 2011; Shahani, Streutker, Dickson, & Stewart, 2007).

Other putative RAADs with published clinical trials in MDD include: (1) Scopolamine (3 i.v. infusions separated by 3–4 days) has shown efficacy compared to placebo in small clinical trials (Drevets, Zarate, & Furey, 2013); (2) Traxoprodil showed efficacy at day 5 following single infusion in a proof of concept study (Preskorn et al., 2008), yet its development was stopped due to incidence of QT prolongation (Machado-Vieira, Henter, & Zarate, 2017); (3) Esketamine, the S enantiomer of ketamine, appears to have RAAD properties following i.v. or i.n. administration in early studies (Canuso et al., 2018; Daly et al., 2018; Singh et al., 2016); (4) Low doses of d-cycloserine, with NMDAR partial agonist effects, were reported to exert RAAD effects in retrospective investigations (Kim, Kushner, Yoon, Anker, & Grant, 2016); (5) Rapastinel (i.v.) has shown efficacy in a proof of concept study (Preskorn et al., 2015); (6) Lanicemine, a low-trapping NMDAR antagonist, has shown efficacy in one phase II study but failed in a second larger clinical trial that may have been complicated by high placebo response rates (Sanacora et al., 2013, 2017; Zarate et al., 2013). Together, these clinical trials provide clear evidence on the prospect of RAADs. However, additional confirmatory clinical trials are still needed to determine the efficacy of these putative RAADs.

As described earlier, the synaptic CSP model would predict that ketamine may have therapeutic effects in many psychiatric disorders with considerable chronic stress component. In fact, pilot trials to date support this hypothesis. Accumulating evidence suggests that ketamine may have independent rapid anti-suicidal effects in depressed patients (Canuso et al., 2018; Grunebaum et al., 2018; Wilkinson et al., 2018). Moreover, pilot evidence suggests potential therapeutic effects of ketamine in treating bipolar depression, PTSD, OCD, GAD, social anxiety disorder (SAD), and substance/alcohol use disorders [(Albott et al., 2018; Diazgranados et al., 2010; Feder et al., 2014; Glue et al., 2017; Ivan Ezquerra-Romano, Lawn, Krupitsky, & Morgan, 2018; Rodriguez et al., 2013; Taylor et al., 2018; Zarate Jr et al., 2012), but also see (Bloch et al., 2012)].

6. Clinical biomarkers of RAADs

To better understand the neurobiology of depression and RAADs, numerous clinical biomarker studies over the past decade capitalized on the RAAD effects of ketamine, its potent effects on prefrontal glutamate neurotransmission, and its robust neuronal remodeling 24 h post infusion. Here, we will briefly review biomarker studies of relevance to the ketamine induced acute glutamate surge (i.e., during infusion) and sustained neuronal remodeling (i.e., 24 h post treatment).

Several lines of evidence have supported the presence of a ketamine induced prefrontal glutamate surge in humans and have associated this surge with the psychotomimetic effects of the drug. Early studies have shown that glutamate release inhibitors would reduce the psychotomimetic effects of ketamine. Later neuroimaging studies have either shown ketamine induced increases in PFC glucose metabolism or blood flow, PFC blood oxygen level dependent (BOLD) signal, or PFC total glutamate level (Anand et al., 2000; Breier, Malhotra, Pinals, Weisenfeld, & Pickar, 1997; Deakin et al., 2008; Javitt et al., 2017; Krystal et al., 2005; Krystal et al., 2010; Milak et al., 2016; Rowland et al., 2005; Stone et al., 2012; Vollenweider et al., 1997; Vollenweider,

Leenders, Oye, Hell, & Angst, 1997). While collectively these studies provide convincing evidence of an acute glutamate surge, most of these studies were in healthy subjects which limits their ability to associate this surge to the RAAD effects. Another main limitation is that these approaches do not distinguish between presynaptic glutamate release and postsynaptic activation. As reviewed earlier, the postsynaptic glutamate activation appears to be the critical process for the RAAD effects. Moreover, recent data using ex vivo (in rats) and in vivo (in humans) ^{13}C MRS suggests that the psychotomimetic effects of ketamine may be due to the decoupling between presynaptic glutamate release and postsynaptic activation (i.e., disruption in communication fidelity across synapses), as evident by increased glutamate cycling combined with inefficient increase in neuroenergetics that are primarily due to postsynaptic activation (i.e., reduction of energy per cycle) [(Chowdhury et al., 2017) & (Abdallah et al. under review)]. If these pilot data were confirmed in future human studies, it will offer a mechanism through which novel drugs may induce RAAD effects without psychotomimetic symptoms, provided that these new agents equally increase presynaptic release and postsynaptic activation.

Further supporting the presence of ketamine induced glutamate surge, recent studies have shown alterations in the binding of metabotropic glutamate receptors subtype 5 (mGluR5) during infusion of ketamine in healthy and depressed subjects (Davis, Holmes, Pietrzak, & Esterlis, 2017; DeLorenzo et al., 2015; Esterlis et al., 2018). Yet, perhaps the most studied approach has been the use of resting state functional MRI. In particular, PFC global brain connectivity (GBC) reduction has been observed in several psychiatric disorders with a strong chronic stress component (Anticevic et al., 2013, 2014, 2015; Cole, Anticevic, Repovs, & Barch, 2011). This observation has led to the hypothesis that GBC may reflect an underlying CSP of reduced PFC synaptic connectivity. Supporting this hypothesis, several studies have demonstrated reduced PFC global connectivity in MDD (Abdallah, Averill, Collins, et al., 2017; Abdallah, Averill, Salas et al., 2017; Murrough et al., 2016; Scheinost et al., 2018; Wang et al., 2014). In addition, human mechanistic studies have provided evidence directly linking glutamate neurotransmission to PFC GBC (Abdallah, Averill, Salas, et al., 2017). Moreover, subanesthetic doses of ketamine have been shown to increase PFC GBC during infusion in healthy individuals, which parallel the hypothesized glutamate surge (Abdallah, Averill, Salas, et al., 2017; Anticevic et al., 2015; Driesen, 2013; Driesen et al., 2013). Consistent with the role of PFC synaptic connectivity in the mechanisms of RAADs, ketamine was found to rapidly normalize PFC GBC abnormalities in MDD patients within 24 h of treatment. These PFC GBC increases were also associated with treatment response (Abdallah, Averill, Collins, et al., 2017). More recently, in a randomized placebo controlled design, it was shown that ketamine increases PFC GBC in MDD during infusion and at 24 h post-treatment. In addition, the amount of PFC GBC increases during ketamine infusion predicted treatment response at 24 h post-treatment (Abdallah et al., 2017). Finally, similar to the preclinical data of ketamine induced increases in hippocampal synaptic connectivity along with reduction in NAc synaptic connectivity (Melo et al., 2015; Reus et al., 2013; Yao et al., 2017), recent pilot human evidence using structural MRI have shown that ketamine significantly increases hippocampal and reduces NAc volumes in MDD patients within 24 h of treatment, particularly in individuals who responded to treatment (Abdallah, Jackowski, Salas, et al., 2017).

7. Current challenges & future directions

The ketamine findings have generated considerable excitement about the promise of a truly novel class of robust and effective RAADs. This excitement was translated into a sizable investment from academia, the pharmaceutical industry, and funding agencies. Hundreds of papers over the past decade have investigated the mechanisms of ketamine and/or its potential therapeutic utility. However, while

preclinical data have extensively investigated ketamine's targets and putative mechanisms, the clinical mechanistic evidence remains lagging.

In particular, to date, we do not have a well-established reproducible biomarker of target engagement (i.e., transient postsynaptic glutamate activation) or target validation (i.e., sustained synaptic remodeling) for the development of RAADs. This issue may have been less problematic for the development of monoaminergic drugs. Following the identification of the in vitro pharmacodynamics of tricyclic antidepressants, the field over the next half a century successfully produced several SAADs, which primarily shared the common in vitro detectable pharmacodynamics of serotonin re-uptake inhibition (SRI). In addition, SRI largely has (1) linear dose response, (2) broad therapeutic window, and (3) relatively stable in vivo pharmacodynamics across administration regimens. In contrast, the development of RAADs has proven more complex; at least two NMDAR antagonists (memantine & CERC-301) and two ampakines (S47445 & ORG 26576) failed in clinical trials and other similar agents have shown promise only in preclinical studies or proof of concept trials. Several challenges may have contributed to the complexity of developing RAADs: (1) While it is evident that blocking NMDAR may induce RAAD effects, only focusing on the in vitro NMDAR antagonism properties appears to have low predictability for the development of new RAADs. (2) Determining the in vivo effects of novel RAAD agents on transient glutamate neurotransmission and sustained synaptic connectivity is critical. However, at this stage, this is only possible in animal studies. The presence of an inverted U-shaped relationship between dose and response further complicates the translatability of preclinical findings. Together, these challenges underscore the need for robust and reproducible biomarkers of prefrontal glutamate activation and synaptic connectivity in humans in vivo. The successful development of these biomarkers would be essential to optimize administration regimens of new RAAD compounds prior to testing them in large expensive trials. Capitalizing on the extensive preclinical and clinical ketamine data, and the swiftness and robustness of its synaptic remodeling and behavioral (psychotomimetic and antidepressant) effects, future studies have a unique opportunity to use ketamine as a tool to establish these synaptic biomarkers that are not only relevant to depression, but also to normal brain function and most neuropsychiatric disorders.

Another opportunity for future studies is that both the chronic stress related synaptic hypoconnectivity and the ketamine induced synaptic hyperconnectivity are reversible within 2 weeks of the intervention, raising mechanistic questions about this individualized homeostatic stable equilibrium of overall synaptic strength. Future studies can capitalize on this notable reversibility to determine the mechanisms underlying this putative homeostatic stable equilibrium of synaptic strength within the context of chronic stress and ketamine treatment. Successful unraveling of these mechanisms may provide information that goes beyond the progress and treatment of depression, and begins to examine the etiology and perhaps cure of depression.

In summary, elegant ketamine studies over the past decade have significantly improved our understanding of the pathophysiology of chronic stress and depression, while unraveling several mechanisms through which transient prefrontal glutamate activation produces rapid restoration of synaptic connectivity along with RAAD effects. Although there is evidence associating glutamate inhibition with antidepressant properties, these data have been primarily limited to SAAD effects. So, is it glutamate inhibition or activation? Within the PFC, chronic stress and depression seem to be associated with high extrasynaptic glutamate level, but overall reduced glutamate neurotransmission as evident by reduction in synaptic connectivity, glutamate cycling, and neuroenergetics. As for the neurobiology of ketamine and other RAADs, it is increasingly evident that transient PFC glutamate postsynaptic activation is a primary underlying mechanism. Yet, it remains to be seen in future studies whether selective

inhibition of extrasynaptic NMDARs would be sufficient to induce sustained synaptic remodeling and robust RAAD effects.

Conflict of interest statement

CGA has served as a consultant and/or on advisory boards for Genentech and Janssen, and editor of *Chronic Stress* for Sage Publications, Inc. GS reports personal consulting fees from Alkermes, Allergan, Biohaven Pharmaceuticals, Eli Lilly and Co., Genentech, Janssen Pharmaceuticals, Lundbeck Research USA, Merck & Co., Naurex, Navitor Pharmaceuticals, Noven Pharmaceuticals, Teva Pharmaceuticals Industries, Taisho Pharmaceutical Co., Takeda Pharmaceutical Co, Sage Pharmaceuticals Inc., Sevier, Valeant Pharmaceuticals, and Vistagen Therapeutics Inc.; grants and research contracts from Eli Lilly and Co., Janssen Pharmaceuticals, Merck & Co., and Sevier and support from Sanofi-Aventis, in the form of free medication for an NIH sponsored study over the last 36 mos. In addition, Dr. Sanacora is a stockholder and holds stock options in Biohaven Pharmaceuticals; and has a patent for Glutamate Modulating Agents in the Treatment of Mental Disorders, U.S. Patent No. 8,778,979 (issued Jul 15, 2014) with royalties paid from Biohaven Pharmaceuticals; GFM is a consultant for Sumitomo Dainippon Pharma Co. Ltd. and UCB Pharma SA, and serves on the Scientific Advisory Board of Elucidata Inc. JHK is a consultant for AbbVie, Inc., Amgen, Astellas Pharma Global Development, Inc., AstraZeneca Pharmaceuticals, Biomedisyn Corporation, Bristol-Myers Squibb, Eli Lilly and Company, Euthymics Bioscience, Inc., Neurovance, Inc., FORUM Pharmaceuticals, Janssen Research & Development, Lundbeck Research USA, Novartis Pharma AG, Otsuka America Pharmaceutical, Inc., Sage Therapeutics, Inc., Sunovion Pharmaceuticals, Inc., and Takeda Industries; is on the Scientific Advisory Board for Lohocla Research Corporation, Mnemosyne Pharmaceuticals, Inc., Naurex, Inc., and Pfizer; is a stockholder in Biohaven Pharmaceuticals; holds stock options in Mnemosyne Pharmaceuticals, Inc.; holds patents for Dopamine and Noradrenergic Reuptake Inhibitors in Treatment of Schizophrenia, U.S. Patent No. 5,447,948 (issued Sep 5, 1995), and Glutamate Modulating Agents in the Treatment of Mental Disorders, U.S. Patent No. 8,778,979 (issued Jul 15, 2014); and filed a patent for Intranasal Administration of Ketamine to Treat Depression. U.S. Application No. 14/197,767 (filed on Mar 5, 2014); U.S. application or Patent Cooperation Treaty international application No. 14/306,382 (filed on Jun 17, 2014).

References

- Abdallah, C. G. (2017). What's the Buzz About Hydroxynorketamine? Is It the History, the Story, the Debate, or the Promise? *Biol Psychiatry* 81, e61–e63.
- Abdallah, C. G., Adams, T. G., Kelmendi, B., Esterlis, I., Sanacora, G., & Krystal, J. H. (2016). Ketamine's mechanism of action: A path to rapid-acting antidepressants. *Depression and Anxiety* 33, 689–697.
- Abdallah, C. G., Averill, C. L., Salas, R., Averill, L. A., Baldwin, P. R., Krystal, J. H., ... Mathalon, D. H. (2017). Prefrontal connectivity and glutamate transmission: Relevance to depression pathophysiology and ketamine treatment. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 2, 566–574.
- Abdallah, C. G., Averill, L. A., Collins, K. A., Geha, P., Schwartz, J., Averill, C., ... Murrough, J. W. (2017). Ketamine treatment and global brain connectivity in major depression. *Neuropsychopharmacology* 42, 1210–1219.
- Abdallah, C. G., Averill, L. A., & Krystal, J. H. (2015). Ketamine as a promising prototype for a new generation of rapid-acting antidepressants. *Annals of the New York Academy of Sciences* 1344, 66–77.
- Abdallah, C. G., Coplan, J. D., Jackowski, A., Sato, J. R., Mao, X., Shungu, D. C., & Mathew, S. J. (2013). A pilot study of hippocampal volume and N-acetylaspartate (NAA) as response biomarkers in riluzole-treated patients with GAD. *European Neuropsychopharmacology* 23, 276–284.
- Abdallah, C. G., Dutta, A., Averill, C. L., McKie, S., Averill, L. A., & Deakin, J. F. (2017). Ketamine, but not the NMDA receptor antagonist lanicemine, increases prefrontal connectivity in depressed patients. *Neuropsychopharmacology* 43, S111.
- Abdallah, C. G., Jackowski, A., Salas, R., Gupta, S., Sato, J. R., Mao, X., ... Mathew, S. J. (2017). The nucleus accumbens and ketamine treatment in major depressive disorder. *Neuropsychopharmacology* 42, 1739–1746.
- Abdallah, C. G., Jackowski, A., Sato, J. R., Mao, X., Kang, G., Cheema, R., ... Shungu, D. C. (2015). Prefrontal cortical GABA abnormalities are associated with reduced hippocampal volume in major depressive disorder. *European Neuropsychopharmacology* 25, 1082–1090.

- Abdallah, C. G., Southwick, S. M., & Krystal, J. H. (2017). Neurobiology of posttraumatic stress disorder (PTSD): A path from novel pathophysiology to innovative therapeutics. *Neuroscience Letters* 649, 130–132.
- Abdallah, C. G., Wrocklage, K. M., Averill, C. L., Akiki, T., Schweinsburg, B., Roy, A., ... Scott, J. C. (2017). Anterior hippocampal dysconnectivity in posttraumatic stress disorder: A dimensional and multimodal approach. *Translational Psychiatry* 7, e1045.
- Adams, T. G., Kelmendi, B., Brake, C. A., Gruner, P., Badour, C. L., & Pittenger, C. (2018). The role of stress in the pathogenesis and maintenance of obsessive-compulsive disorder. *Chronic Stress* 2, 2470547018758043.
- Akiki, T. J., Averill, C. L., Wrocklage, K. M., Schweinsburg, B., Scott, J. C., Martini, B., ... Abdallah, C. G. (2017). The association of PTSD symptom severity with localized Hippocampus and amygdala abnormalities. *Chronic Stress* 1, 2470547017724069.
- Albott, C. S., Lim, K. O., Forbes, M. K., Erbes, C., Tye, S. J., Grabowski, J. G., ... Shiroma, P. R. (2018). Efficacy, safety, and durability of repeated ketamine infusions for comorbid posttraumatic stress disorder and treatment-resistant depression. *Journal of Clinical Psychiatry* 79.
- Aleksandrova, L. R., Wang, Y. T., & Phillips, A. G. (2017). Hydroxynorketamine: Implications for the NMDA receptor hypothesis of ketamine's antidepressant action. *Chronic Stress* 1, 2470547017743511.
- Anand, A., Charney, D. S., Oren, D. A., Berman, R. M., Hu, X. S., Cappiello, A., & Krystal, J. H. (2000). Attenuation of the neuropsychiatric effects of ketamine with lamotrigine: Support for hyperglutamatergic effects of N-methyl-D-aspartate receptor antagonists. *Archives of General Psychiatry* 57, 270–276.
- Anticevic, A., Brumbaugh, M. S., Winkler, A. M., Lombardo, L. E., Barrett, J., Corlett, P. R., ... Glahn, D. C. (2013). Global prefrontal and fronto-amygdala dysconnectivity in bipolar I disorder with psychosis history. *Biological Psychiatry* 73, 565–573.
- Anticevic, A., Corlett, P. R., Cole, M. W., Savic, A., Gancsos, M., Tang, Y., ... Krystal, J. H. (2015). N-methyl-D-aspartate receptor antagonist effects on prefrontal cortical connectivity better model early than chronic schizophrenia. *Biological Psychiatry* 77, 569–580.
- Anticevic, A., Hu, S., Zhang, S., Savic, A., Billingslea, E., Wasylink, S., ... Pittenger, C. (2014). Global resting-state functional magnetic resonance imaging analysis identifies frontal cortex, striatal, and cerebellar dysconnectivity in obsessive-compulsive disorder. *Biological Psychiatry* 75, 595–605.
- Autry, A. E., Adachi, M., Nosyreva, E., Na, E. S., Los, M. F., Cheng, P. F., ... Monteggia, L. M. (2011). NMDA receptor blockade at rest triggers rapid behavioural antidepressant responses. *Nature* 475, 91–95. <https://doi.org/10.1038/nature10130>.
- Averill, L. A., Abdallah, C. G., Pietrzak, R. H., Averill, C. L., Southwick, S. M., Krystal, J. H., & Harpaz-Rotem, I. (2017). Combat exposure severity is associated with reduced cortical thickness in combat veterans: A preliminary report. *Chronic Stress* 1, 2470547017724714.
- Averill, L. A., Purohit, P., Averill, C. L., Boesl, M. A., Krystal, J. H., & Abdallah, C. G. (2017). Glutamate dysregulation and glutamatergic therapeutics for PTSD: Evidence from human studies. *Neuroscience Letters* 649, 147–155.
- Ban, T. A. (2006). The role of serendipity in drug discovery. *Dialogues in Clinical Neuroscience* 8, 335–344.
- Banasr, M., Chowdhury, G. M., Terwilliger, R., Newton, S. S., Duman, R. S., Behar, K. L., & Sanacora, G. (2010). Glial pathology in an animal model of depression: Reversal of stress-induced cellular, metabolic and behavioral deficits by the glutamate-modulating drug riluzole. *Molecular Psychiatry* 15, 501–511.
- Berman, R. M., Cappiello, A., Anand, A., Oren, D. A., Heninger, G. R., Charney, D. S., & Krystal, J. H. (2000). Antidepressant effects of ketamine in depressed patients. *Biological Psychiatry* 47, 351–354.
- Bloch, M. H., Wasylink, S., Landeros-Weisenberger, A., Panza, K. E., Billingslea, E., Leckman, J. F., ... Pittenger, C. (2012). Effects of ketamine in treatment-refractory obsessive-compulsive disorder. *Biological Psychiatry* 72, 964–970.
- Bobo, W. V., Vande Voort, J. L., Croarkin, P. E., Leung, J. G., Tye, S. J., & Frye, M. A. (2016). Ketamine for treatment-resistant unipolar and bipolar major depression: Critical review and implications for clinical practice. *Depression and Anxiety* 33, 698–710.
- Breier, A., Malhotra, A. K., Pinals, D. A., Weisenfeld, N. I., & Pickar, D. (1997). Association of ketamine-induced psychosis with focal activation of the prefrontal cortex in healthy volunteers. *The American Journal of Psychiatry* 154, 805–811.
- Bremner, J. D., Randall, P., Scott, T. M., Bronen, R. A., Seibyl, J. P., Southwick, S. M., ... Innis, R. B. (1995). MRI-based measurement of hippocampal volume in patients with combat-related posttraumatic stress disorder. *The American Journal of Psychiatry* 152, 973–981.
- Campioni, M. R., Xu, M., & McGehee, D. S. (2009). Stress-induced changes in nucleus accumbens glutamate synaptic plasticity. *Journal of Neurophysiology* 101, 3192–3198.
- Can, A., Zanos, P., Moaddel, R., Kang, H. J., Dossou, K. S., Wainer, I. W., ... Gould, T. D. (2016). Effects of ketamine and ketamine metabolites on evoked striatal dopamine release, dopamine receptors, and monoamine transporters. *The Journal of Pharmacology and Experimental Therapeutics* 359, 159–170.
- Canuso, C. M., Singh, J. B., Fedgchin, M., Alphas, L., Lane, R., Lim, P., ... Drevets, W. C. (2018). Efficacy and safety of intranasal esketamine for the rapid reduction of symptoms of depression and suicidality in patients at imminent risk for suicide: results of a double-blind, randomized, placebo-controlled study. *The American Journal of Psychiatry*. <https://doi.org/10.1176/appi.ajp.2018.17060720>.
- Chaudhury, D., Walsh, J. J., Friedman, A. K., Juarez, B., Ku, S. M., Koo, J. W., ... Han, M. H. (2013). Rapid regulation of depression-related behaviours by control of midbrain dopamine neurons. *Nature* 493, 532–536.
- Chowdhury, G. M., Banasr, M., de Graaf, R. A., Rothman, D. L., Behar, K. L., & Sanacora, G. (2008). Chronic riluzole treatment increases glucose metabolism in rat prefrontal cortex and hippocampus. *Journal of Cerebral Blood Flow and Metabolism* 28, 1892–1897.
- Chowdhury, G. M., Behar, K. L., Cho, W., Thomas, M. A., Rothman, D. L., & Sanacora, G. (2012). 1H-[13C]-nuclear magnetic resonance spectroscopy measures of ketamine's effect on amino acid neurotransmitter metabolism. *Biological Psychiatry* 71, 1022–1025.
- Chowdhury, G. M., Zhang, J., Thomas, M., Banasr, M., Ma, X., Pittman, B., ... Sanacora, G. (2017). Transiently increased glutamate cycling in rat PFC is associated with rapid onset of antidepressant-like effects. *Mol Psychiatry/Molecular Psychiatry* 22, 120–126.
- Christoffel, D. J., Golden, S. A., Dumitriu, D., Robison, A. J., Janssen, W. G., Ahn, H. F., ... Russo, S. J. (2011). IkappaB kinase regulates social defeat stress-induced synaptic and behavioral plasticity. *The Journal of Neuroscience* 31, 314–321.
- Christoffel, D. J., Golden, S. A., Heshmati, M., Graham, A., Birnbaum, S., Neve, R. L., ... Russo, S. J. (2012). Effects of inhibitor of kappaB kinase activity in the nucleus accumbens on emotional behavior. *Neuropsychopharmacology* 37, 2615–2623.
- Cohen, M. L., Chan, S. L., Way, W. L., & Trevor, A. J. (1973). Distribution in the brain and metabolism of ketamine in the rat after intravenous administration. *Anesthesiology* 39, 370–376.
- Cole, M. W., Anticevic, A., Repovs, G., & Barch, D. (2011). Variable global dysconnectivity and individual differences in schizophrenia. *Biological Psychiatry* 70, 43–50.
- Coplan, J. D., Lu, D., El Sehamy, A. M., Tang, C., Jackowski, A. P., Abdallah, C. G., ... Gorman, J. M. (2018). Early life stress associated with increased striatal N-acetyl-aspartate (NAA): Cerebrospinal fluid (CSF) corticotropin-releasing factor (CRF) concentrations, hippocampal volume, body mass and behavioral correlates. *Chronic Stress* 2 (In Press).
- Cottrell, A., Warren, K., Ayres, R., Weinstock, P., Kumar, V., & Gillatt, D. (2008). The destruction of the lower urinary tract by ketamine abuse: A new syndrome? *BJU International* 102, 1178–1179 (author reply 1179).
- Crane, G. E. (1959). Cyloserine as an antidepressant agent. *The American Journal of Psychiatry* Vol. 115, 1025–1026.
- Daly, E. J., Singh, J. B., Fedgchin, M., Cooper, K., Lim, P., Shelton, R. C., ... Drevets, W. C. (2018). Efficacy and safety of intranasal esketamine adjunctive to oral antidepressant therapy in treatment-resistant depression: A randomized clinical trial. *JAMA Psychiatry* 75, 139–148.
- Daskalakis, N. P., & Binder, E. B. (2015). Schizophrenia in the spectrum of gene-stress interactions: The FKBP5 example. *Schizophrenia Bulletin* 41, 323–329.
- Davis, M. T., Holmes, S. E., Pietrzak, R. H., & Esterlis, I. (2017). Neurobiology of chronic stress-related psychiatric disorders: Evidence from molecular imaging studies. *Chronic Stress* 1, 2470547017710916.
- Deakin, J. F., Lees, J., McKie, S., Hallak, J. E., Williams, S. R., & Dursun, S. M. (2008). Glutamate and the neural basis of the subjective effects of ketamine: A pharmacomagnetic resonance imaging study. *Archives of General Psychiatry* 65, 154–164.
- DeLorenzo, C., DellaGioia, N., Bloch, M., Sanacora, G., Nabulsi, N., Abdallah, C., ... Esterlis, I. (2015). In vivo ketamine-induced changes in [(11)C]ABP688 binding to metabotropic glutamate receptor subtype 5. *Biological Psychiatry* 77, 266–275.
- Deutschenbaur, L., Beck, J., Kiyhankhadi, A., Muhlhauser, M., Borgwardt, S., Walter, M., ... Lang, U. E. (2016). Role of calcium, glutamate and NMDA in major depression and therapeutic application. *Progress in Neuro-Psychopharmacology & Biological Psychiatry* 64, 325–333.
- Diazgranados, N., Ibrahim, L., Brutsche, N. E., Newberg, A., Kronstein, P., Khalife, S., ... Zarate, C. A., Jr. (2010). A randomized add-on trial of an N-methyl-D-aspartate antagonist in treatment-resistant bipolar depression. *Archives of General Psychiatry* 67, 793–802.
- Drevets, W. C., Zarate, C. A., Jr., & Furey, M. L. (2013). Antidepressant effects of the muscarinic cholinergic receptor antagonist scopolamine: A review. *Biological Psychiatry* 73, 1156–1163.
- Driesen, N. R., McCarthy, G., Bhagwagar, Z., Bloch, M., Calhoun, V., D'Souza, D. C., ... Krystal, J. H. (2013b). Relationship of resting brain hyperconnectivity and schizophrenia-like symptoms produced by the NMDA receptor antagonist ketamine in humans. *Molecular Psychiatry* 18, 1199–1204.
- Driesen, N. R., McCarthy, G., Bhagwagar, Z., Bloch, M. H., Calhoun, V. D., D'Souza, D. C., ... Krystal, J. H. (2013a). The impact of NMDA receptor blockade on human working memory-related prefrontal function and connectivity. *Neuropsychopharmacology* 38, 2613–2622.
- D'Sa, C., & Duman, R. S. (2002). Antidepressants and neuroplasticity. *Bipolar Disorders* 4, 183–194.
- Duman, R. S., & Aghajanian, G. K. (2012). Synaptic dysfunction in depression: Potential therapeutic targets. *Science* 338, 68–72.
- Duman, R. S., Aghajanian, G. K., Sanacora, G., & Krystal, J. H. (2016). Synaptic plasticity and depression: New insights from stress and rapid-acting antidepressants. *Nature Medicine* 22, 238–249.
- Duman, R. S., Heninger, G. R., & Nestler, E. J. (1997). A molecular and cellular theory of depression. *Archives of General Psychiatry* 54, 597–606.
- Esterlis, I., DellaGioia, N., Pietrzak, R. H., Matuskey, D., Nabulsi, N., Abdallah, C. G., ... DeLorenzo, C. (2018). Ketamine-induced reduction in mGluR5 availability is associated with an antidepressant response: an [(11)C]ABP688 and PET imaging study in depression. *Molecular Psychiatry* 23, 824–832.
- Feder, A., Parides, M. K., Murrough, J. W., Perez, A. M., Morgan, J. E., Saxena, S., ... Charney, D. S. (2014). Efficacy of intravenous ketamine for treatment of chronic posttraumatic stress disorder: A randomized clinical trial. *JAMA Psychiatry* 71, 681–688.
- Gall, C. M., & Isackson, P. J. (1989). Limbic seizures increase neuronal production of messenger RNA for nerve growth factor. *Science* 245, 758–761.
- Ghosal, S., Bang, E., Yue, W., Hare, B. D., Lepack, A. E., Girgenti, M. J., & Duman, R. S. (2018). Activity-dependent brain-derived neurotrophic factor release is required for the rapid antidepressant actions of scopolamine. *Biological Psychiatry* 83, 29–37.
- Glue, P., Medicott, N. J., Harland, S., Neehoff, S., Anderson-Fahey, B., Le Nedelec, M., ... McNaughton, N. (2017). Ketamine's dose-related effects on anxiety symptoms in patients with treatment refractory anxiety disorders. *Journal of Psychopharmacology* 31, 1302–1305.

- Goddard, A. W. (2017). The neurobiology of panic: A chronic stress disorder. *Chronic Stress* 1, 2470547017736038.
- Grunebaum, M. F., Galfalvy, H. C., Choo, T. H., Keilp, J. G., Moitra, V. K., Parris, M. S., ... Mann, J. J. (2018). Ketamine for rapid reduction of suicidal thoughts in major depression: A midazolam-controlled randomized clinical trial. *The American Journal of Psychiatry* 175, 327–335.
- Hardingham, G. E., & Bading, H. (2010). Synaptic versus extrasynaptic NMDA receptor signalling: Implications for neurodegenerative disorders. *Nature Reviews. Neuroscience* 11, 682–696.
- Hare, B., Ghosal, S., & Duman, R. (2017). Rapid acting antidepressants in chronic stress models: Molecular and cellular mechanisms. *Chronic Stress* 1, 1–16.
- Haukvik, U. K., Westlye, L. T., Mørch-Johnsen, L., Jørgensen, K. N., Lange, E. H., Dale, A. M., ... Agartz, I. (2015). In vivo hippocampal subfield volumes in schizophrenia and bipolar disorder. *Biological Psychiatry* 77, 581–588.
- Homayoun, H., & Moghaddam, B. (2007). NMDA receptor hypofunction produces opposite effects on prefrontal cortex interneurons and pyramidal neurons. *The Journal of Neuroscience* 27, 11496–11500.
- Ivan Ezquerro-Romano, I., Lawn, W., Krupitsky, E., & Morgan, C. J. A. (2018). Ketamine for the treatment of addiction: Evidence and potential mechanisms. *Neuropharmacology*. <https://doi.org/10.1016/j.neuropharm.2018.01.017>.
- Javitt, D. C. (2004). Glutamate as a therapeutic target in psychiatric disorders. *Molecular Psychiatry* 9, 984–997, 979.
- Javitt, D. C., Carter, C. S., Krystal, J. H., Kantrowitz, J. T., Girgis, R. R., Kegeles, L. S., ... Lieberman, J. A. (2018). Utility of imaging-based biomarkers for glutamate-targeted drug development in psychotic disorders: A randomized clinical trial. *JAMA Psychiatry* 75, 11–19.
- Jourdi, H., Hsu, Y. T., Zhou, M., Qin, Q., Bi, X., & Baudry, M. (2009). Positive AMPA receptor modulation rapidly stimulates BDNF release and increases dendritic mRNA translation. *The Journal of Neuroscience* 29, 8688–8697.
- Karasawa, J., Shimazaki, T., Kawashima, N., & Chaki, S. (2005). AMPA receptor stimulation mediates the antidepressant-like effect of a group II metabotropic glutamate receptor antagonist. *Brain Research* 1042, 92–98.
- Kassem, M. S., Lagopoulos, J., Stait-Gardner, T., Price, W. S., Chohan, T. W., Arnold, J. C., ... Bennett, M. R. (2013). Stress-induced grey matter loss determined by MRI is primarily due to loss of dendrites and their synapses. *Molecular Neurobiology* 47, 645–661.
- Kempton, M. J., Salvador, Z., Munafo, M. R., Geddes, J. R., Simmons, A., Frangou, S., & Williams, S. C. (2011). Structural neuroimaging studies in major depressive disorder. Meta-analysis and comparison with bipolar disorder. *Archives of General Psychiatry* 68, 675–690.
- Kim, S. W., Kushner, M. G., Yoon, G., Anker, J., & Grant, J. E. (2016). Low-dose d-cycloserine for depression? *The Journal of Clinical Psychiatry* 77, e1007.
- Klein, D. F. (2008). The loss of serendipity in psychopharmacology. *JAMA* 299, 1063–1065.
- Kokkinou, M., Ashok, A. H., & Howes, O. D. (2018). The effects of ketamine on dopaminergic function: Meta-analysis and review of the implications for neuropsychiatric disorders. *Molecular Psychiatry* 23, 59–69.
- Krishnan, V., & Nestler, E. J. (2008). The molecular neurobiology of depression. *Nature* 455, 894–902.
- Krystal, J. H., Abdallah, C. G., Averill, L. A., Kelmendi, B., Harpaz-Rotem, I., Sanacora, G., ... Duman, R. S. (2017). Synaptic loss and the pathophysiology of PTSD: Implications for ketamine as a prototype novel therapeutic. *Current Psychiatry Reports* 19, 74.
- Krystal, J. H., Abi-Saab, W., Perry, E., D'Souza, D. C., Liu, N., Gueorgieva, R., ... Breier, A. (2005). Preliminary evidence of attenuation of the disruptive effects of the NMDA glutamate receptor antagonist, ketamine, on working memory by pretreatment with the group II metabotropic glutamate receptor agonist, LY354740, in healthy human subjects. *Psychopharmacology* 179, 303–309.
- Krystal, J. H., Mathew, S. J., D'Souza, D. C., Garakani, A., Gunduz-Bruce, H., & Charney, D. S. (2010). Potential psychiatric applications of metabotropic glutamate receptor agonists and antagonists. *CNS Drugs* 24, 669–693.
- Krystal, J. H., Sanacora, G., Blumberg, H., Anand, A., Charney, D. S., Marek, G., ... Mason, G. F. (2002). Glutamate and GABA systems as targets for novel antidepressant and mood-stabilizing treatments. *Molecular Psychiatry* 7(Suppl. 1), S71–S80.
- Kwako, L. E., & Koob, G. F. (2017). Neuroclinical framework for the role of stress in addiction. *Chronic Stress* 1, 2470547017698140.
- Kwon, J. S., Shin, Y. W., Kim, C. W., Kim, Y. I., Youn, T., Han, M. H., ... Kim, J. J. (2003). Similarity and disparity of obsessive-compulsive disorder and schizophrenia in MR volumetric abnormalities of the hippocampus-amygdala complex. *Journal of Neurology, Neurosurgery, and Psychiatry* 74, 962–964.
- Lapidus, K. A., Levitch, C. F., Perez, A. M., Brallier, J. W., Parides, M. K., Soleimani, L., ... Murrough, J. W. (2014). A randomized controlled trial of intranasal ketamine in major depressive disorder. *Biological Psychiatry* 76, 970–976.
- Lauterborn, J. C., Lynch, G., Vanderklis, P., Arai, A., & Gall, C. M. (2000). Positive modulation of AMPA receptors increases neurotrophin expression by hippocampal and cortical neurons. *The Journal of Neuroscience* 20, 8–21.
- Lepack, A. E., Bang, E., Lee, B., Dwyer, J. M., & Duman, R. S. (2016). Fast-acting antidepressants rapidly stimulate ERK signaling and BDNF release in primary neuronal cultures. *Neuropharmacology* 111, 242–252.
- Lepack, A. E., Fuchikami, M., Dwyer, J. M., Banas, M., & Duman, R. S. (2015). BDNF release is required for the behavioral actions of ketamine. *The International Journal of Neuropsychopharmacology* 18.
- Li, N., He, X., Zhang, Y., Qi, X., Li, H., Zhu, X., & He, S. (2011). Brain-derived neurotrophic factor signalling mediates antidepressant effects of lamotrigine. *The International Journal of Neuropsychopharmacology* 14, 1091–1098.
- Li, N., Lee, B., Liu, R. J., Banas, M., Dwyer, J. M., Iwata, M., ... Duman, R. S. (2010). mTOR-dependent synapse formation underlies the rapid antidepressant effects of NMDA antagonists. *Science* 329, 959–964.
- Li, S. X., Han, Y., Xu, L. Z., Yuan, K., Zhang, R. X., Sun, C. Y., ... Lu, L. (2018). Uncoupling DAPK1 from NMDA receptor GluN2B subunit exerts rapid antidepressant-like effects. *Molecular Psychiatry* 23, 597–608.
- Li, X., Tizzano, J. P., Griffey, K., Clay, M., Lindstrom, T., & Skolnick, P. (2001). Antidepressant-like actions of an AMPA receptor potentiator (LY392098). *Neuropharmacology* 40, 1028–1033.
- Liu, R. J., Duman, C., Kato, T., Hare, B., Lopresto, D., Bang, E., ... Duman, R. S. (2017). GLYX-13 produces rapid antidepressant responses with key synaptic and behavioral effects distinct from ketamine. *Neuropsychopharmacology* 42, 1231–1242.
- Liu, R. J., Lee, F. S., Li, X. Y., Bambico, F., Duman, R. S., & Aghajanian, G. K. (2012). Brain-derived neurotrophic factor Val66Met allele impairs basal and ketamine-stimulated synaptogenesis in prefrontal cortex. *Biological Psychiatry* 71, 996–1005.
- Machado-Vieira, R., Henter, I. D., & Zarate, C. A., Jr. (2017). New targets for rapid antidepressant action. *Progress in Neurobiology* 152, 21–37.
- Maeng, L. Y., & Milad, M. R. (2017). Post-traumatic stress disorder: The relationship between the fear response and chronic stress. *Chronic Stress* 1, 2470547017713297.
- Maeng, S., Zarate, C. A., Jr., Du, J., Schloesser, R. J., McCammon, J., Chen, G., & Manji, H. K. (2008). Cellular mechanisms underlying the antidepressant effects of ketamine: Role of alpha-amino-3-hydroxy-5-methylisoxazole-4-propionic acid receptors. *Biological Psychiatry* 63, 349–352.
- Manji, H. K., Drevets, W. C., & Charney, D. S. (2001). The cellular neurobiology of depression. *Nature Medicine* 7, 541–547.
- Mathew, S. J., Gueorgieva, R., Brandt, C., Fava, M., & Sanacora, G. (2017). A randomized, double-blind, placebo-controlled, sequential parallel comparison design trial of adjunctive riluzole for treatment-resistant major depressive disorder. *Neuropsychopharmacology* 42, 2567–2574.
- Matosin, N., Cruceanu, C., & Binder, E. B. (2017). Preclinical and clinical evidence of DNA methylation changes in response to trauma and chronic stress. *Chronic Stress* 1, 2470547017710764.
- McEwen, B. S. (1999). Stress and hippocampal plasticity. *Annual Review of Neuroscience* 22, 105–122.
- McEwen, B. S. (2004). Protection and damage from acute and chronic stress: Allostasis and allostatic overload and relevance to the pathophysiology of psychiatric disorders. *Annals of the New York Academy of Sciences* 1032, 1–7.
- McGirr, A., Berlim, M. T., Bond, D. J., Fleck, M. P., Yatham, L. N., & Lam, R. W. (2015). A systematic review and meta-analysis of randomized, double-blind, placebo-controlled trials of ketamine in the rapid treatment of major depressive episodes. *Psychological Medicine* 45, 693–704.
- Melo, A., Kokras, N., Dalla, C., Ferreira, C., Ventura-Silva, A. P., Sousa, N., & Pego, J. M. (2015). The positive effect on ketamine as a priming adjuvant in antidepressant treatment. *Translational Psychiatry* 5, e573.
- Milak, M. S., Proper, C. J., Mulhern, S. T., Parter, A. L., Kegeles, L. S., Ogden, R. T., ... Mann, J. J. (2016). A pilot in vivo proton magnetic resonance spectroscopy study of amino acid neurotransmitter response to ketamine treatment of major depressive disorder. *Molecular Psychiatry* 21, 320–327.
- Miller, O. H., Moran, J. T., & Hall, B. J. (2016). Two cellular hypotheses explaining the initiation of ketamine's antidepressant actions: Direct inhibition and disinhibition. *Neuropharmacology* 100, 17–26.
- Moaddel, R., Sanghvi, M., Dossou, K. S., Ramamoorthy, A., Green, C., Bupp, J., ... Wainer, I. W. (2015). The distribution and clearance of (2S,6S)-hydroxynorketamine, an active ketamine metabolite, in Wistar rats. *Pharmacology Research & Perspectives* 3, e00157.
- Moghaddam, B. (1993). Stress preferentially increases extraneuronal levels of excitatory amino acids in the prefrontal cortex: Comparison to hippocampus and basal ganglia. *Journal of Neurochemistry* 60, 1650–1657.
- Moghaddam, B., Adams, B., Verma, A., & Daly, D. (1997). Activation of glutamatergic neurotransmission by ketamine: A novel step in the pathway from NMDA receptor blockade to dopaminergic and cognitive disruptions associated with the prefrontal cortex. *The Journal of Neuroscience* 17, 2921–2927.
- Morgan, C. J., Curran, H. V., & Independent Scientific Committee on Drugs (2012). Ketamine use: A review. *Addiction* 107, 27–38.
- Muhammad, A., Carroll, C., & Kolb, B. (2012). Stress during development alters dendritic morphology in the nucleus accumbens and prefrontal cortex. *Neuroscience* 216, 103–109.
- Murrough, J. W., Abdallah, C. G., Anticevic, A., Collins, K. A., Geha, P., Averill, L. A., ... Charney, D. S. (2016). Reduced global functional connectivity of the medial prefrontal cortex in major depressive disorder. *Human Brain Mapping* 37, 3214–3223.
- Murrough, J. W., Abdallah, C. G., & Mathew, S. J. (2017). Targeting glutamate signalling in depression: Progress and prospects. *Nature Reviews. Drug Discovery* 16, 472–486.
- Murrough, J. W., Iosifescu, D. V., Chang, L. C., Al Jurdi, R. K., Green, C. E., Perez, A. M., ... Mathew, S. J. (2013). Antidepressant efficacy of ketamine in treatment-resistant major depression: A two-site randomized controlled trial. *The American Journal of Psychiatry* 170, 1134–1142.
- Nibuya, M., Morinobu, S., & Duman, R. S. (1995). Regulation of BDNF and trkB mRNA in rat brain by chronic electroconvulsive seizure and antidepressant drug treatments. *The Journal of Neuroscience* 15, 7539–7547.
- Noppers, I. M., Niesters, M., Aarts, L. P., Bauer, M. C., Drewes, A. M., Dahan, A., & Sarton, E. Y. (2011). Drug-induced liver injury following a repeated course of ketamine treatment for chronic pain in CRPS type 1 patients: a report of 3 cases. *Pain* 152, 2173–2178.
- Paoletti, P., Bellone, C., & Zhou, Q. (2013). NMDA receptor subunit diversity: Impact on receptor properties, synaptic plasticity and disease. *Nature Reviews. Neuroscience* 14, 383–400.
- Patriquin, M. A., & Mathew, S. J. (2017). The neurobiological mechanisms of generalized anxiety disorder and chronic stress. *Chronic Stress* 1, 2470547017703993.

- Patterson, S. L., Grover, L. M., Schwartzkroin, P. A., & Bothwell, M. (1992). Neurotrophin expression in rat hippocampal slices: A stimulus paradigm inducing LTP in CA1 evokes increases in BDNF and NT-3 mRNAs. *Neuron* 9, 1081–1088.
- Paul, R. K., Singh, N. S., Khader, M., Moaddel, R., Sanghvi, M., Green, C. E., ... Wainer, I. W. (2014). (R,S)-Ketamine metabolites (R,S)-norketamine and (2S,6S)-hydroxynorketamine increase the mammalian target of rapamycin function. *Anesthesiology* 121, 149–159.
- Pham, T. H., Defaix, C., Xu, X., Deng, S. X., Fabresse, N., Alvarez, J. C., ... Gardier, A. M. (2017). Common neurotransmission recruited in (R,S)-ketamine and (2R,6R)-hydroxynorketamine-induced sustained antidepressant-like effects. *Biological Psychiatry*. <https://doi.org/10.1016/j.biopsych.2017.10.020>.
- Pietrzak, R. H., Averill, L. A., Abdallah, C. G., Neumeister, A., Krystal, J. H., Levy, I., & Harpaz-Rotem, I. (2015). Amygdala-hippocampal volume and the phenotypic heterogeneity of posttraumatic stress disorder: A cross-sectional study. *JAMA Psychiatry* 72, 396–398.
- Popoli, M., Yan, Z., McEwen, B. S., & Sanacora, G. (2012). The stressed synapse: The impact of stress and glucocorticoids on glutamate transmission. *Nature Reviews. Neuroscience* 13, 22–37.
- Prescott, A., Sheth, C., Legarreta, M., Renshaw, P. F., McGlade, E., & Yurgelun-Todd, D. (2018). Altered cortical gamma-aminobutyric acid in female veterans with suicidal behavior: Sex differences and clinical correlates. *Chronic Stress* 2, 2470547018768771.
- Preskorn, S., Macaluso, M., Mehra, D. O., Zammit, G., Moskal, J. R., Burch, R. M., & Group, G. C. S. (2015). Randomized proof of concept trial of GLYX-13, an N-methyl-D-aspartate receptor glycine site partial agonist, in major depressive disorder nonresponsive to a previous antidepressant agent. *Journal of Psychiatric Practice* 21, 140–149.
- Preskorn, S. H., Baker, B., Kolluri, S., Menniti, F. S., Krams, M., & Landen, J. W. (2008). An innovative design to establish proof of concept of the antidepressant effects of the NR2B subunit selective N-methyl-D-aspartate antagonist, CP-101,606, in patients with treatment-refractory major depressive disorder. *Journal of Clinical Psychopharmacology* 28, 631–637.
- Reus, G. Z., Abelaira, H. M., dos Santos, M. A., Carlessi, A. S., Tomaz, D. B., Neotti, M. V., ... Quevedo, J. (2013). Ketamine and imipramine in the nucleus accumbens regulate histone deacetylation induced by maternal deprivation and are critical for associated behaviors. *Behavioural Brain Research* 256, 451–456.
- Rodriguez, C. I., Kegeles, L. S., Levinson, A., Feng, T., Marcus, S. M., Vermes, D., ... Simpson, H. B. (2013). Randomized controlled crossover trial of ketamine in obsessive-compulsive disorder: Proof-of-concept. *Neuropsychopharmacology* 38, 2475–2483.
- Rothman, D. L., De Feyter, H. M., de Graaf, R. A., Mason, G. F., & Behar, K. L. (2011). ¹³C MRS studies of neuroenergetics and neurotransmitter cycling in humans. *NMR in Biomedicine* 24, 943–957.
- Rowland, L. M., Bustillo, J. R., Mullins, P. G., Jung, R. E., Lenroot, R., Landgraf, E., ... Brooks, W. M. (2005). Effects of ketamine on anterior cingulate glutamate metabolism in healthy humans: A 4-T proton MRS study. *The American Journal of Psychiatry* 162, 394–396.
- Russo, S. J., & Nestler, E. J. (2013). The brain reward circuitry in mood disorders. *Nature Reviews. Neuroscience* 14, 609–625.
- Sanacora, G., & Banasr, M. (2013). From pathophysiology to novel antidepressant drugs: Glial contributions to the pathology and treatment of mood disorders. *Biological Psychiatry* 73, 1172–1179.
- Sanacora, G., Frye, M. A., McDonald, W., Mathew, S. J., Turner, M. S., Schatzberg, A. F., ... American Psychiatric Association Council of Research Task Force on Novel, B., & Treatments (2017). A consensus statement on the use of ketamine in the treatment of mood disorders. *JAMA Psychiatry* 74, 399–405.
- Sanacora, G., Johnson, M. R., Khan, A., Atkinson, S. D., Riesenberger, R. R., Schronen, J. P., ... Pathak, S. (2017). Adjunctive lamicemine (AZD6765) in patients with major depressive disorder and history of inadequate response to antidepressants: A randomized, placebo-controlled study. *Neuropsychopharmacology* 42, 844–853.
- Sanacora, G., Smith, M. A., Pathak, S., Su, H. L., Boeijinga, P. H., McCarthy, D. J., & Quirk, M. C. (2014). Lamicemine: A low-trapping NMDA channel blocker produces sustained antidepressant efficacy with minimal psychotomimetic adverse effects. *Molecular Psychiatry* 19, 978–985.
- Sanacora, G., Treccani, G., & Popoli, M. (2012). Towards a glutamate hypothesis of depression: An emerging frontier of neuropsychopharmacology for mood disorders. *Neuropharmacology* 62, 63–77.
- Scheinost, D., Holmes, S. E., DellaGioia, N., Schleifer, C., Matuskey, D., Abdallah, C. G., ... Esterlis, I. (2018). Multimodal investigation of network level effects using intrinsic functional connectivity, anatomical covariance, and structure-to-function correlations in unmedicated major depressive disorder. *Neuropsychopharmacology* 43, 1119–1127.
- Schobel, S. A., Chaudhury, N. H., Khan, U. A., Paniagua, B., Styner, M. A., Asllani, I., ... Small, S. A. (2013). Imaging patients with psychosis and a mouse model establishes a spreading pattern of hippocampal dysfunction and implicates glutamate as a driver. *Neuron* 78, 81–93.
- Schwarcz, R., Wu, H.-Q., Zanos, P., & Gould, T. (2017). Endogenous kynurenic acid mediates ketamine- and (2R,6R)-hydroxynorketamine-induced increases in extracellular glutamate and antidepressant actions. *Neuropsychopharmacology* 43, S111.
- Shahani, R., Streutker, C., Dickson, B., & Stewart, R. J. (2007). Ketamine-associated ulcerative cystitis: A new clinical entity. *Urology* 69, 810–812.
- Sheline, Y. I., Wang, P. W., Gado, M. H., Csernansky, J. G., & Vannier, M. W. (1996). Hippocampal atrophy in recurrent major depression. *Proceedings of the National Academy of Sciences of the United States of America* 93, 3908–3913.
- Sheth, C., McGlade, E., & Yurgelun-Todd, D. (2017). Chronic stress in adolescents and its neurobiological and psychopathological consequences: An RDoC perspective. *Chronic Stress* 1, 2470547017715645.
- Singh, J. B., Fedgchin, M., Daly, E., Xi, L., Melman, C., De Bruecker, G., ... Van Nueten, L. (2016). Intravenous esketamine in adult treatment-resistant depression: A double-blind, double-randomization, placebo-controlled study. *Biological Psychiatry* 80, 424–431.
- Singh, J. B., Fedgchin, M., Daly, E. J., De Boer, P., Cooper, K., Lim, P., ... Van Nueten, L. (2016). A double-blind, randomized, placebo-controlled, dose-frequency study of intravenous ketamine in patients with treatment-resistant depression. *The American Journal of Psychiatry* 173, 816–826.
- Skolnick, P., Lauer, R. T., Popik, P., Nowak, G., Paul, I. A., & Trullas, R. (1996). Adaptation of N-methyl-D-aspartate (NMDA) receptors following antidepressant treatment: Implications for the pharmacotherapy of depression. *Pharmacopsychiatry* 29, 23–26.
- Solmi, M., Veronese, N., Zaninotto, L., van der Loos, M. L., Gao, K., Schaffer, A., ... Correll, C. U. (2016). Lamotrigine compared to placebo and other agents with antidepressant activity in patients with unipolar and bipolar depression: A comprehensive meta-analysis of efficacy and safety outcomes in short-term trials. *CNS Spectrums* 21, 403–418.
- Stone, J. M., Dietrich, C., Edden, R., Mehta, M. A., De Simoni, S., Reed, L. J., ... Barker, G. J. (2012). Ketamine effects on brain GABA and glutamate levels with 1H-MRS: Relationship to ketamine-induced psychopathology. *Molecular Psychiatry* 17, 664–665.
- Syed, S. A., & Nemeroff, C. B. (2017). Early life stress, mood, and anxiety disorders. *Chronic Stress* 1, 2470547017694461.
- Tang, H., Kukral, D., Li, Y. W., Fronheiser, M., Malone, H., Pena, A., ... Luo, F. (2018). Mapping the central effects of (+/-)-ketamine and traxoprodil using pharmacological magnetic resonance imaging in awake rats. *Journal of Psychopharmacology* 32, 146–155.
- Taylor, J. H., Landerer-Weisenberger, A., Coughlin, C., Mulqueen, J., Johnson, J. A., Gabriel, D., ... Bloch, M. H. (2018). Ketamine for social anxiety disorder: A randomized, placebo-controlled crossover trial. *Neuropsychopharmacology* 43, 325–333.
- Trullas, R., & Skolnick, P. (1990). Functional antagonists at the NMDA receptor complex exhibit antidepressant actions. *European Journal of Pharmacology* 185, 1–10.
- Voleti, B., Navarria, A., Liu, R. J., Banasr, M., Li, N., Terwilliger, R., ... Duman, R. S. (2013). Scopamine rapidly increases mammalian target of rapamycin complex 1 signaling, synaptogenesis, and antidepressant behavioral responses. *Biological Psychiatry* 74, 742–749.
- Vollenweider, F. X., Leenders, K. L., Oye, I., Hell, D., & Angst, J. (1997). Differential psychopathology and patterns of cerebral glucose utilisation produced by (S)- and (R)-ketamine in healthy volunteers using positron emission tomography (PET). *European Neuropsychopharmacology* 7, 25–38.
- Vollenweider, F. X., Leenders, K. L., Scharfetter, C., Antonini, A., Maguire, P., Missimer, J., & Angst, J. (1997). Metabolic hyperfrontality and psychopathology in the ketamine model of psychosis using positron emission tomography (PET) and [¹⁸F] fluorodeoxyglucose (FDG). *European Neuropsychopharmacology* 7, 9–24.
- Walsh, J. J., Friedman, A. K., Sun, H., Heller, E. A., Ku, S. M., Juarez, B., ... Han, M. H. (2014). Stress and CRF gate neural activation of BDNF in the mesolimbic reward pathway. *Nature Neuroscience* 17, 27–29.
- Wang, L., Dai, Z., Peng, H., Tan, L., Ding, Y., He, Z., ... Li, L. (2014). Overlapping and segregated resting-state functional connectivity in patients with major depressive disorder with and without childhood neglect. *Human Brain Mapping* 35, 1154–1166.
- Warren, B. L., Sial, O. K., Alcantara, L. F., Greenwood, M. A., Brewer, J. S., Rozofsky, J. P., ... Bolanos-Guzman, C. A. (2014). Altered gene expression and spine density in nucleus accumbens of adolescent and adult mice exposed to emotional and physical stress. *Developmental Neuroscience* 36, 250–260.
- Wilkinson, S. T., Ballard, E. D., Bloch, M. H., Mathew, S. J., Murrough, J. W., Feder, A., ... Sanacora, G. (2018). The effect of a single dose of intravenous ketamine on suicidal ideation: A systematic review and individual participant data meta-analysis. *The American Journal of Psychiatry* 175, 150–158.
- Wohleb, E. S., Wu, M., Gerhard, D. M., Taylor, S. R., Picciotto, M. R., Alreja, M., & Duman, R. S. (2016). GABA interneurons mediate the rapid antidepressant-like effects of scopamine. *The Journal of Clinical Investigation* 126, 2482–2494.
- Wook Koo, J., Labonte, B., Engmann, O., Calipari, E. S., Juarez, B., Lorsch, Z., ... Nestler, E. J. (2016). Essential role of mesolimbic brain-derived neurotrophic factor in chronic social stress-induced depressive behaviors. *Biological Psychiatry* 80, 469–478.
- Wrocklage, K. M., Averill, L. A., Cobb Scott, J., Averill, C. L., Schweinsburg, B., Trejo, M., ... Abdallah, C. G. (2017). Cortical thickness reduction in combat exposed U.S. veterans with and without PTSD. *European Neuropsychopharmacology* 27, 515–525.
- Yang, Y., Cui, Y., Sang, K., Dong, Y., Ni, Z., Ma, S., & Hu, H. (2018). Ketamine blocks bursting in the lateral habenula to rapidly relieve depression. *Nature* 554, 317–322.
- Yao, N., Skiteva, O., Zhang, X., Svenningsson, P., & Chergui, K. (2017). Ketamine and its metabolite (2R,6R)-hydroxynorketamine induce lasting alterations in glutamatergic synaptic plasticity in the mesolimbic circuit. *Molecular Psychiatry* <https://doi.org/10.1038/mp.2017.239>.
- Yuen, E. Y., Liu, W., Karatsoreos, I. N., Feng, J., McEwen, B. S., & Yan, Z. (2009). Acute stress enhances glutamatergic transmission in prefrontal cortex and facilitates working memory. *Proceedings of the National Academy of Sciences of the United States of America* 106, 14075–14079.
- Yuen, E. Y., Liu, W., Karatsoreos, I. N., Ren, Y., Feng, J., McEwen, B. S., & Yan, Z. (2011). Mechanisms for acute stress-induced enhancement of glutamatergic transmission and working memory. *Molecular Psychiatry* 16, 156–170.
- Yuen, E. Y., Wei, J., Liu, W., Zhong, P., Li, X., & Yan, Z. (2012). Repeated stress causes cognitive impairment by suppressing glutamate receptor expression and function in prefrontal cortex. *Neuron* 73, 962–977.
- Zafra, F., Castren, E., Thoenen, H., & Lindholm, D. (1991). Interplay between glutamate and gamma-aminobutyric acid transmitter systems in the physiological regulation of brain-derived neurotrophic factor and nerve growth factor synthesis in hippocampal

- neurons. *Proceedings of the National Academy of Sciences of the United States of America* 88, 10037–10041.
- Zanos, P., Moaddel, R., Morris, P. J., Georgiou, P., Fischell, J., Elmer, G. I., ... Gould, T. D. (2016). NMDAR inhibition-independent antidepressant actions of ketamine metabolites. *Nature* 533, 481–486.
- Zarate, C. A., Jr., Brutsche, N. E., Ibrahim, L., Franco-Chaves, J., Diazgranados, N., Cravchik, A., ... Luckenbaugh, D. A. (2012). Replication of ketamine's antidepressant efficacy in bipolar depression: A randomized controlled add-on trial. *Biological Psychiatry* 71, 939–946.
- Zarate, C. A., Jr., Mathews, D., Ibrahim, L., Chaves, J. F., Marquardt, C., Ukoh, I., ... Luckenbaugh, D. A. (2013). A randomized trial of a low-trapping nonselective N-methyl-D-aspartate channel blocker in major depression. *Biological Psychiatry* 74, 257–264.
- Zarate, C. A., Jr., Singh, J. B., Carlson, P. J., Brutsche, N. E., Ameli, R., Luckenbaugh, D. A., ... Manji, H. K. (2006). A randomized trial of an N-methyl-D-aspartate antagonist in treatment-resistant major depression. *Archives of General Psychiatry* 63, 856–864.